Chapter 8. Conformal Mappings

Conformal equivalence, Schwarz lemma and automorphisms, Riemann mapping theorem, Schwarz–Christoffel

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Overview

This chapter is geometric and "global": we study mapping properties of holomorphic functions, motivated by the question

> *Given two open sets U and V in $\mathbb{C}$, does there exist a holomorphic bijection between them?*

(The inverse will automatically be holomorphic.) The most fruitful choice is $V = \mathbb{D}$: *what conditions on Ω guarantee a holomorphic bijection $\Omega \to \mathbb{D}$?* (For $V = \mathbb{C}$ only $U = \mathbb{C}$ is possible.) We first give explicit examples (the upper half-plane maps to the disc by a fractional linear transformation; compositions of elementary maps) and apply them to the Dirichlet problem. Then the Schwarz lemma determines the automorphisms of the disc and of the upper half-plane. The heart of the chapter is the *Riemann mapping theorem*: Ω is conformally equivalent to $\mathbb{D}$ whenever it is simply connected and not all of $\mathbb{C}$, with no assumption on $\partial\Omega$. For polygons the map is given by the *Schwarz–Christoffel formula*; for rectangles it involves elliptic integrals, which lead to the doubly periodic functions of Chapter 9.

1 Conformal equivalence and examples

A bijective holomorphic $f : U \to V$ is a **conformal map** or **biholomorphism**; then U and V are **conformally equivalent** (**biholomorphic**).

Proposition 1.1

If $f : U \to V$ is holomorphic and injective, then $f'(z) \neq 0$ for all $z \in U$. In particular, the inverse of f defined on its range is holomorphic, and thus the inverse of a conformal map is also holomorphic.

Proof. Suppose $f'(z_0) = 0$. Then $f(z) - f(z_0) = a(z - z_0)^k + G(z)$ near z_0, with $a \neq 0$, $k \geq 2$, and G vanishing to order $k + 1$ at z_0. For small w write $f(z) - f(z_0) - w = F(z) + G(z)$ with $F(z) = a(z - z_0)^k - w$. On a small circle about z_0, $|G| < |F|$, and F has at least two zeros inside, so by Rouché's theorem $f(z) - f(z_0) - w$ has at least two zeros there. Since $f'(z) \neq 0$ for $z \neq z_0$ near z_0 (zeros of f' are isolated), these roots are distinct, so f is not injective.

Let $g = f^{-1}$ on the range V, $w_0 = f(z_0)$, $w = f(z)$ close to w_0, $w \neq w_0$. Then

$$\frac{g(w) - g(w_0)}{w - w_0} = \frac{1}{\frac{w - w_0}{g(w) - g(w_0)}} = \frac{1}{\frac{f(z) - f(z_0)}{z - z_0}}.$$

Since $f'(z_0) \neq 0$, letting $z \to z_0$ shows g is holomorphic at w_0 with $g'(w_0) = 1/f'(g(w_0))$. ∎

So U, V are conformally equivalent iff there are holomorphic $f : U \to V$, $g : V \to U$ with $g(f(z)) = z$ and $f(g(w)) = w$. (Some authors call f conformal when merely $f' \neq 0$, e.g. z^2 on $\mathbb{C} - \{0\}$, which is not injective. The condition $f'(z) \neq 0$ means f is a local bijection, and such a map preserves angles between curves.)

1.1 The disc and upper half-plane

The **upper half-plane** is $\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im}(z) > 0\}$. Let

$$F(z) = \frac{i - z}{i + z} \qquad \text{and} \qquad G(w) = i\frac{1 - w}{1 + w}.$$

Theorem 1.2

The map $F : \mathbb{H} \to \mathbb{D}$ is a conformal map with inverse $G : \mathbb{D} \to \mathbb{H}$.

Proof. Both maps are holomorphic in their domains. Points of $\mathbb{H}$ are closer to i than to $-i$, so $|F(z)| < 1$. For $w = u + iv \in \mathbb{D}$,

$$\operatorname{Im}(G(w)) = \operatorname{Re}\Big(\frac{1-u-iv}{1+u+iv}\Big) = \operatorname{Re}\frac{(1-u-iv)(1+u-iv)}{(1+u)^2+v^2} = \frac{1-u^2-v^2}{(1+u)^2+v^2} > 0.$$

Finally

$$F(G(w)) = \frac{i - i\frac{1-w}{1+w}}{i + i\frac{1-w}{1+w}} = \frac{1+w-1+w}{1+w+1-w} = w,$$

and similarly $G(F(z)) = z$. ∎

F is holomorphic on $\mathbb{C}$ except at $-i$, hence continuous on $\mathbb{R} = \partial\mathbb{H}$, and $|F(x)| = 1$ for real x. With $x = \tan t$, $t \in (-\pi/2, \pi/2)$,

$$F(x) = \frac{1-x^2}{1+x^2} + i\frac{2x}{1+x^2} = \cos 2t + i \sin 2t = e^{2it},$$

so F maps $\mathbb{R}$ onto the circle minus -1, traversed from -1 first through the lower half; -1 corresponds to the point at infinity of $\mathbb{H}$.

Remark

Maps $z \mapsto \frac{az+b}{cz+d}$ ($a, b, c, d \in \mathbb{C}$, denominator not a multiple of the numerator) are **fractional linear transformations**; they reappear as the automorphisms of the disc and of $\mathbb{H}$ (Theorems 2.2, 2.4).

1.2 Further examples

Example 1

Translations and dilations. For $h \in \mathbb{C}$, $z \mapsto z+h$ is a conformal map of $\mathbb{C}$ (inverse $w \mapsto w-h$); for real h it is also a conformal map of $\mathbb{H}$. For $c \neq 0$, $z \mapsto cz$ is conformal on $\mathbb{C}$ (inverse $w \mapsto c^{-1}w$): a rotation if $c = e^{i\varphi}$, a dilation if $c > 0$, a dilation by $|c|$ followed by rotation by π if $c < 0$.

Example 2

For a positive integer n, $z \mapsto z^n$ is conformal from the sector $\{0 < \arg z < \pi/n\}$ to $\mathbb{H}$, with inverse $w \mapsto w^{1/n}$ (principal branch). More generally, for $0 < \alpha < 2$, $f(z) = z^\alpha$ maps $\mathbb{H}$ to the sector $S = \{0 < \arg w < \alpha\pi\}$: with the branch obtained by deleting the positive real axis and $z = re^{i\theta}$, $0 < \theta < \pi$, $f(z) = |z|^\alpha e^{i\alpha\theta}$; the inverse is $w^{1/\alpha}$ with $0 < \arg w < \alpha\pi$. Composing with Example 1, $\mathbb{H}$ maps conformally onto any infinite sector. On the boundary: as x goes from $-\infty$ to 0, $f(x)$ goes from $\infty e^{i\alpha\pi}$ to 0 along $\arg z = \alpha\pi$; as x goes from 0 to ∞, $f(x)$ goes from 0 to ∞ along $\mathbb{R}$.

Example 3

$f(z) = \frac{1+z}{1-z}$ maps the upper half-disc $\{|z| < 1,\ y > 0\}$ conformally onto the first quadrant

$\{u > 0, v > 0\}$: for $z = x + iy$,

$$f(z) = \frac{1 - (x^2 + y^2)}{(1-x)^2 + y^2} + i\frac{2y}{(1-x)^2 + y^2}.$$

The inverse $g(w) = \frac{w-1}{w+1}$ is holomorphic in the first quadrant, maps it into $\mathbb{D}$ (the distance from w to -1 exceeds that to 1) with $\operatorname{Im} g(w) > 0$. On the boundary, for $z = e^{i\theta}$, $f(z) = \frac{e^{-i\theta/2}+e^{i\theta/2}}{e^{-i\theta/2}-e^{i\theta/2}} = \frac{i}{\tan(\theta/2)}$, so as θ goes from 0 to π, $f(e^{i\theta})$ travels down the imaginary axis from ∞ to 0; f maps $(-1, 1)$ increasingly onto $(0, \infty)$, $f(0) = 1$.

Example 4

$z \mapsto \log z$, the branch obtained by deleting the negative imaginary axis, maps $\mathbb{H}$ onto the strip $\{w = u + iv : 0 < v < \pi\}$: for $z = re^{i\theta}$, $-\pi/2 < \theta < 3\pi/2$, $\log z = \log r + i\theta$. The inverse is $w \mapsto e^w$. As x goes from $-\infty$ to 0, $f(x)$ goes from $\infty + i\pi$ to $-\infty + i\pi$; as x goes from 0 to ∞, $f(x)$ goes from $-\infty$ to ∞ along $\mathbb{R}$.

Example 5

$z \mapsto \log z$ also maps the half-disc $\{|z| < 1, y > 0\}$ onto the half-strip $\{u < 0,\ 0 < v < \pi\}$: $(0, 1)$ goes to $(-\infty, 0)$, the upper half-circle from 1 to -1 to the segment from 0 to $i\pi$, and $(-1, 0)$ to the half-line from $i\pi$ to $-\infty + i\pi$.

Example 6

$f(z) = e^{iz}$ maps the half-strip $\{-\pi/2 < x < \pi/2,\ y > 0\}$ onto the half-disc $\{|w| < 1,\ u > 0\}$, since $e^{iz} = e^{-y}e^{ix}$. The boundary goes: from $\pi/2 + i\infty$ to $\pi/2$, f goes from 0 to i; from $\pi/2$ to $-\pi/2$, f goes from i to $-i$ along the half-circle; from $-\pi/2$ to $-\pi/2 + i\infty$, from $-i$ back to 0. (This is closely related to the inverse of Example 5.)

Example 7

$f(z) = -\frac{1}{2}(z+1/z)$ maps the half-disc $\{|z| < 1, y > 0\}$ conformally onto $\mathbb{H}$. On the boundary: as x goes from 0 to 1, $f(x)$ goes from $-\infty$ to -1; on the half-circle $z = e^{i\theta}$, $f(z) = -\cos\theta$ runs from -1 to 1 as θ goes from 0 to π; and as x goes from -1 to 0, $f(x)$ goes from 1 to ∞.

Example 8

$f(z) = \sin z$ maps the half-strip $\{-\pi/2 < x < \pi/2,\ y > 0\}$ conformally onto $\mathbb{H}$. With $\zeta = e^{iz}$,

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i} = -\frac{1}{2}\Big(i\zeta + \frac{1}{i\zeta}\Big),$$

so f is the map of Example 6, followed by multiplication by i (rotation by $\pi/2$, giving the upper half-disc), followed by Example 7. As z goes from $-\pi/2 + i\infty$ to $-\pi/2$, f goes from $-\infty$ to -1; on $[-\pi/2, \pi/2]$, f runs over $[-1, 1]$; from $\pi/2$ to $\pi/2 + i\infty$, f goes from 1 to ∞.

1.3 The Dirichlet problem in a strip

The **Dirichlet problem** in Ω is

$$\triangle u = 0 \text{ in } \Omega, \qquad u = f \text{ on } \partial\Omega, \tag{1}$$

$\triangle = \partial^2/\partial x^2 + \partial^2/\partial y^2$, f given on $\partial\Omega$. For the disc and $\mathbb{H}$ it is solved by convolution with Poisson kernels. We solve it for a strip using conformal maps and the solution in the disc.

Lemma 1.3

Let V and U be open sets in $\mathbb{C}$ and $F : V \to U$ a holomorphic function. If $u : U \to \mathbb{C}$ is a harmonic function, then $u \circ F$ is harmonic on V.

Proof. The question is local, so take U a disc. There is G holomorphic on U with $\operatorname{Re} G = u$ (every harmonic function in a disc is the real part of a holomorphic function). Then $u \circ F = \operatorname{Re}(G \circ F)$ with $G \circ F$ holomorphic, so $u \circ F$ is harmonic. ∎

Let $\Omega = \{x + iy : x \in \mathbb{R},\ 0 < y < 1\}$, with boundary data $u(x,0) = f_0(x)$, $u(x,1) = f_1(x)$, f_0, f_1 continuous and vanishing at infinity. Transfer the problem to $\mathbb{D}$ by

$$F(w) = \frac{1}{\pi} \log\Big(i\frac{1-w}{1+w}\Big) : \mathbb{D} \to \Omega, \qquad G(z) = \frac{i - e^{\pi z}}{i + e^{\pi z}} : \Omega \to \mathbb{D},$$

compositions of maps above, conformal and mutually inverse. F maps the lower half-circle onto $i+\mathbb{R}$ and the upper half-circle onto $\mathbb{R}$: as φ goes from $-\pi$ to 0, $F(e^{i\varphi})$ goes from $i+\infty$ to $i-\infty$, and as φ goes from 0 to π, $F(e^{i\varphi})$ goes from $-\infty$ to ∞. Define $\tilde{f}_1(\varphi) = f_1(F(e^{i\varphi}) - i)$ for $-\pi < \varphi < 0$ and $\tilde{f}_0(\varphi) = f_0(F(e^{i\varphi}))$ for $0 < \varphi < \pi$; since f_0, f_1 vanish at infinity, $\tilde{f}$ (equal to $\tilde{f}_1$, $\tilde{f}_0$ on the two half-circles and 0 at $\varphi = \pm\pi, 0$) is continuous on the circle. The solution in the disc is the Poisson integral, for $w = re^{i\theta}$,

$$\tilde{u}(w) = \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(\theta-\varphi)\tilde{f}(\varphi)\,d\varphi = \frac{1}{2\pi}\int_{-\pi}^{0} P_r(\theta-\varphi)\tilde{f}_1(\varphi)\,d\varphi + \frac{1}{2\pi}\int_{0}^{\pi} P_r(\theta-\varphi)\tilde{f}_0(\varphi)\,d\varphi,$$

$P_r(\gamma) = \frac{1-r^2}{1-2r\cos\gamma+r^2}$ the Poisson kernel. By Lemma 1.3, $u(z) = \tilde{u}(G(z))$ is harmonic in the strip, with the correct boundary values. For $z = iy$, $0 < y < 1$, with $re^{i\theta} = G(iy)$, a change of variables gives

$$\frac{1}{2\pi}\int_{0}^{\pi} P_r(\theta-\varphi)\tilde{f}_0(\varphi)\,d\varphi = \frac{\sin\pi y}{2}\int_{-\infty}^{\infty}\frac{f_0(t)}{\cosh\pi t - \cos\pi y}\,dt, \qquad \frac{1}{2\pi}\int_{-\pi}^{0} P_r(\theta-\varphi)\tilde{f}_1(\varphi)\,d\varphi = \frac{\sin\pi y}{2}\int_{-\infty}^{\infty}\frac{f_1(t}{\cosh\pi t +}$$

A solution vanishing at infinity is unique, so translating the boundary data by x translates the solution, and

$$u(x,y) = \frac{\sin\pi y}{2}\Big(\int_{-\infty}^{\infty}\frac{f_0(x-t)}{\cosh\pi t - \cos\pi y}\,dt + \int_{-\infty}^{\infty}\frac{f_1(x-t)}{\cosh\pi t + \cos\pi y}\,dt\Big),$$

convolutions with f_0 and f_1. At $y = 1/2$ the kernel is $1/\cosh\pi t$, its own Fourier transform (Example 3 in Section 2 of Chapter 3).

Remarks about the Dirichlet problem. If $F : \mathbb{D} \to \Omega$ is conformal and extends to a continuous bijection $\partial\mathbb{D} \to \partial\Omega$, then for f on $\partial\Omega$, $\tilde{f} = f \circ F$ is defined on the circle, the Poisson integral $\tilde{u}(re^{i\theta}) = \frac{1}{2\pi}\int_0^{2\pi} P_r(\theta-\varphi)\tilde{f}(e^{i\varphi})\,d\varphi$ solves the problem in the disc, and $u = \tilde{u} \circ F^{-1}$ is expected to solve (1). This requires:

(i) existence of a conformal map $\Phi = F^{-1} : \Omega \to \mathbb{D}$, settled in general by the Riemann mapping theorem;

(ii) its extension to a continuous bijection of the boundaries, which needs some regularity of $\partial\Omega$ (treated for polygons in Section 4). Riemann reversed the implication: to find Φ with $\Phi(z_0) = 0$, write $\Phi(z) = (z - z_0)e^{H(z)}$; $|\Phi| = 1$ on $\partial\Omega$ means $u = \operatorname{Re} H$ solves the Dirichlet problem with $u(z) = \log(1/|z - z_0|)$ on $\partial\Omega$ (u is the Green's function with source z_0). This approach requires showing Φ is a bijection, some boundary regularity, and solving the Dirichlet problem (Riemann proposed the "Dirichlet principle", whose implementation has difficulties). We prove the existence in general by different methods in Section 3.

2 The Schwarz lemma; automorphisms of the disc and upper half-plane

A **rotation** is a map $z \mapsto cz$ with $|c| = 1$, $c = e^{i\varphi}$, φ the angle of rotation (defined modulo 2π).

Lemma 2.1 (Schwarz lemma)

Let $f : \mathbb{D} \to \mathbb{D}$ be holomorphic with $f(0) = 0$. Then (i) $|f(z)| \le |z|$ for all $z \in \mathbb{D}$.
(ii) If for some $z_0 \neq 0$ we have $|f(z_0)| = |z_0|$, then f is a rotation.
(iii) $|f'(0)| \le 1$, and if equality holds, then f is a rotation.

Proof. Expand $f(z) = a_0 + a_1 z + a_2 z^2 + \cdots$ in $\mathbb{D}$; $f(0) = 0$ gives $a_0 = 0$, so $f(z)/z$ has a removable singularity at 0 and is holomorphic in $\mathbb{D}$. If $|z| = r < 1$, since $|f(z)| \le 1$, $|f(z)/z| \le 1/r$, and by the maximum modulus principle this holds for $|z| \le r$. Letting $r \to 1$ gives (i). For (ii), $f(z)/z$ attains its maximum in the interior of $\mathbb{D}$, so it is a constant c, and $|c| = 1$ (evaluate at z_0). For (iii), $g(z) = f(z)/z$ satisfies $|g| \le 1$ on $\mathbb{D}$ and $g(0) = \lim_{z\to 0} \frac{f(z)-f(0)}{z} = f'(0)$; if $|f'(0)| = 1$, then $|g(0)| = 1$, so g is constant by the maximum principle and $f(z) = cz$, $|c| = 1$. ∎

2.1 Automorphisms of the disc

A conformal map of Ω to itself is an **automorphism** of Ω; they form a group $\mathrm{Aut}(\Omega)$ under composition, with $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$. Rotations $r_\theta : z \mapsto e^{i\theta} z$ are automorphisms of $\mathbb{D}$ (inverse $r_{-\theta}$). More interesting are

$$\psi_\alpha(z) = \frac{\alpha - z}{1 - \overline{\alpha} z}, \qquad \alpha \in \mathbb{C},\ |\alpha| < 1.$$

ψ_α is holomorphic in $\mathbb{D}$. If $z = e^{i\theta}$, $\psi_\alpha(e^{i\theta}) = \frac{\alpha - e^{i\theta}}{e^{i\theta}(e^{-i\theta} - \overline{\alpha})} = -e^{-i\theta}\frac{w}{\overline{w}}$ with $w = \alpha - e^{i\theta}$, so $|\psi_\alpha| = 1$ on the circle and $|\psi_\alpha| < 1$ in $\mathbb{D}$ by the maximum modulus principle. Moreover

$$(\psi_\alpha \circ \psi_\alpha)(z) = \frac{\alpha - \frac{\alpha - z}{1 - \overline{\alpha} z}}{1 - \overline{\alpha}\frac{\alpha - z}{1 - \overline{\alpha} z}} = \frac{\alpha - |\alpha|^2 z - \alpha + z}{1 - \overline{\alpha} z - |\alpha|^2 + \overline{\alpha} z} = \frac{(1 - |\alpha|^2) z}{1 - |\alpha|^2} = z,$$

so ψ_α is its own inverse; and it interchanges 0 and α: $\psi_\alpha(0) = \alpha$, $\psi_\alpha(\alpha) = 0$.

Theorem 2.2

If f is an automorphism of the disc, then there exist $\theta \in \mathbb{R}$ and $\alpha \in \mathbb{D}$ such that

$$f(z) = e^{i\theta} \frac{\alpha - z}{1 - \overline{\alpha} z}.$$

Proof. There is a unique $\alpha \in \mathbb{D}$ with $f(\alpha) = 0$. Let $g = f \circ \psi_\alpha$; then $g(0) = 0$ and by the Schwarz lemma

$$|g(z)| \le |z| \qquad \text{for all } z \in \mathbb{D}. \tag{2}$$

Also $g^{-1}(0) = 0$, so $|g^{-1}(w)| \le |w|$; with $w = g(z)$,

$$|z| \le |g(z)| \qquad \text{for all } z \in \mathbb{D}. \tag{3}$$

By (2), (3), $|g(z)| = |z|$, so $g(z) = e^{i\theta} z$ by the Schwarz lemma. Replacing z by $\psi_\alpha(z)$ and using $\psi_\alpha \circ \psi_\alpha = \mathrm{id}$, $f(z) = e^{i\theta}\psi_\alpha(z)$. ∎

Corollary 2.3

The only automorphisms of the unit disc that fix the origin are the rotations.

The group $\mathrm{Aut}(\mathbb{D})$ acts *transitively*: given $\alpha, \beta \in \mathbb{D}$, $\psi_\beta \circ \psi_\alpha$ maps α to β. $\mathrm{Aut}(\mathbb{D})$ is "almost" isomorphic to the matrix group $SU(1,1)$ of 2×2 complex matrices preserving the hermitian form $\langle Z, W \rangle = z_1\overline{w_1} - z_2\overline{w_2}$ on $\mathbb{C}^2 \times \mathbb{C}^2$.

2.2 Automorphisms of the upper half-plane

With $F : \mathbb{H} \to \mathbb{D}$ from Section 1, define $\Gamma : \mathrm{Aut}(\mathbb{D}) \to \mathrm{Aut}(\mathbb{H})$ by "conjugation by F": $\Gamma(\varphi) = F^{-1} \circ \varphi \circ F$. Γ is a bijection with inverse $\Gamma^{-1}(\psi) = F \circ \psi \circ F^{-1}$, and it preserves composition:

$$\Gamma(\varphi_1 \circ \varphi_2) = F^{-1} \circ \varphi_1 \circ F \circ F^{-1} \circ \varphi_2 \circ F = \Gamma(\varphi_1) \circ \Gamma(\varphi_2).$$

So $\mathrm{Aut}(\mathbb{D}) \cong \mathrm{Aut}(\mathbb{H})$. Let $SL_2(\mathbb{R}) = \left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R},\ \det(M) = ad - bc = 1 \right\}$, the **special linear group**, and for $M \in SL_2(\mathbb{R})$ let $f_M(z) = \frac{az+b}{cz+d}$.

Theorem 2.4

Every automorphism of $\mathbb{H}$ takes the form f_M for some $M \in SL_2(\mathbb{R})$. Conversely, every map of this form is an automorphism of $\mathbb{H}$.

Proof. Write $G = SL_2(\mathbb{R})$.

Step 1. If $M \in G$, f_M maps $\mathbb{H}$ to itself:

$$\mathrm{Im}(f_M(z)) = \frac{(ad-bc)\,\mathrm{Im}(z)}{|cz+d|^2} = \frac{\mathrm{Im}(z)}{|cz+d|^2} > 0 \qquad (z \in \mathbb{H}). \tag{4}$$

Step 2. $f_M \circ f_{M'} = f_{MM'}$ (a straightforward calculation). Hence each f_M is an automorphism, with holomorphic inverse $f_{M^{-1}}$: $(f_M \circ f_{M^{-1}})(z) = f_I(z) = z$.

Step 3. G acts transitively on $\mathbb{H}$; it suffices to map any $z \in \mathbb{H}$ to i. With $d = 0$ in (4), $\mathrm{Im}(f_M(z)) = \mathrm{Im}(z)/|cz|^2$, and we choose $c \in \mathbb{R}$ making this 1; let $M_1 = \begin{pmatrix} 0 & -c^{-1} \\ c & 0 \end{pmatrix}$, so $f_{M_1}(z)$ has imaginary part 1. Then translate by $M_2 = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$, $b \in \mathbb{R}$, to bring it to i; $M = M_2 M_1$ works.

Step 4. For $\theta \in \mathbb{R}$, $M_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \in G$, and $F \circ f_{M_\theta} \circ F^{-1}$ is the rotation of angle -2θ in the disc, since $F \circ f_{M_\theta} = e^{-2i\theta} F(z)$ (easily verified).

Step 5. Let $f \in \mathrm{Aut}(\mathbb{H})$ with $f(\beta) = i$, and $N \in G$ with $f_N(i) = \beta$. Then $g = f \circ f_N$ fixes i, so $F \circ g \circ F^{-1}$ is an automorphism of $\mathbb{D}$ fixing 0, hence a rotation; by Step 4, $F \circ g \circ F^{-1} = F \circ f_{M_\theta} \circ F^{-1}$ for some θ. So $g = f_{M_\theta}$ and $f = f_{M_\theta N^{-1}}$. ∎

M and $-M$ give the same f_M; identifying them gives the **projective special linear group** $PSL_2(\mathbb{R})$, which is isomorphic to $\mathrm{Aut}(\mathbb{H})$.

3 The Riemann mapping theorem

3.1 Necessary conditions and statement of the theorem

If $\Omega = \mathbb{C}$ there is no conformal map $F : \Omega \to \mathbb{D}$ (by Liouville F would be constant); Ω must be connected (as $\mathbb{D}$ is) and simply connected (as $\mathbb{D}$ is). Remarkably, these conditions suffice. A subset of $\mathbb{C}$ is **proper** if it is non-empty and not all of $\mathbb{C}$.

Theorem 3.1 (Riemann mapping theorem)

Suppose Ω is proper and simply connected. If $z_0 \in \Omega$, then there exists a unique conformal map $F : \Omega \to \mathbb{D}$ such that $F(z_0) = 0$ and $F'(z_0) > 0$.

Corollary 3.2

Any two proper simply connected open subsets in $\mathbb{C}$ are conformally equivalent.

The corollary follows using $\mathbb{D}$ as an intermediate step. *Uniqueness*: if F, G satisfy the conditions, $H = F \circ G^{-1}$ is an automorphism of $\mathbb{D}$ fixing 0, so $H(z) = e^{i\theta}z$; $H'(0) > 0$ forces $e^{i\theta} = 1$, so $F = G$. For existence, consider all injective holomorphic $f : \Omega \to \mathbb{D}$ with $f(z_0) = 0$, and choose one with $|f'(z_0)|$ as large as possible; it will fill out $\mathbb{D}$. We need to extract limits from sequences of functions.

3.2 Montel's theorem

A family $\mathcal{F}$ of holomorphic functions on Ω is **normal** if every sequence in $\mathcal{F}$ has a subsequence converging uniformly on every compact subset of Ω (the limit need not be in $\mathcal{F}$). $\mathcal{F}$ is **uniformly bounded on compact subsets** of Ω if for each compact $K \subset \Omega$ there is $B > 0$ with $|f(z)| \le B$ for $z \in K$, $f \in \mathcal{F}$. $\mathcal{F}$ is **equicontinuous** on a compact K if for every $\epsilon > 0$ there is $\delta > 0$ such that $z, w \in K$, $|z - w| < \delta$ imply $|f(z) - f(w)| < \epsilon$ for all $f \in \mathcal{F}$. (E.g. differentiable functions on $[0, 1]$ with uniformly bounded derivatives are equicontinuous; $f_n(x) = x^n$ on $[0, 1]$ are not.)

Theorem 3.3 (Montel)

Suppose $\mathcal{F}$ is a family of holomorphic functions on Ω that is uniformly bounded on compact subsets of Ω. Then:
(i) $\mathcal{F}$ is equicontinuous on every compact subset of Ω.
(ii) $\mathcal{F}$ is a normal family.

Part (i) uses the Cauchy integral formula and contrasts with the real case ($\sin(nx)$ on $(0, 1)$ is uniformly bounded but not equicontinuous, with no convergent subsequence on any subinterval). Part (ii) uses only uniform boundedness and equicontinuity (the Arzelà–Ascoli theorem). A sequence $\{K_\ell\}_{\ell=1}^{\infty}$ of compact subsets of Ω is an **exhaustion** if (a) K_ℓ is contained in the interior of $K_{\ell+1}$ for all ℓ, and (b) any compact $K \subset \Omega$ is contained in some K_ℓ; in particular $\Omega = \bigcup_\ell K_\ell$.

Lemma 3.4

Any open set Ω in the complex plane has an exhaustion.

Proof. If Ω is bounded, let K_ℓ be the set of points of Ω at distance $\ge 1/\ell$ from $\partial\Omega$. If Ω is unbounded, require also $|z| \le \ell$ for $z \in K_\ell$. ■

Proof of Montel's theorem. (i) Let $K \subset \Omega$ be compact and $r > 0$ with $3r < d(K, \partial\Omega)$, so $D_{3r}(z) \subset \Omega$ for $z \in K$. Let $z, w \in K$ with $|z - w| < r$ and γ the boundary circle of $D_{2r}(w)$. By Cauchy's integral formula

$$f(z) - f(w) = \frac{1}{2\pi i}\int_\gamma f(\zeta)\Big[\frac{1}{\zeta - z} - \frac{1}{\zeta - w}\Big]d\zeta, \qquad \Big|\frac{1}{\zeta - z} - \frac{1}{\zeta - w}\Big| = \frac{|z - w|}{|\zeta - z||\zeta - w|} \le \frac{|z - w|}{r^2},$$

since $\zeta \in \gamma$ and $|z - w| < r$. Hence $|f(z) - f(w)| \le \frac{1}{2\pi} 2\pi(2r)\frac{B|z-w|}{r^2}$, B the uniform bound for $\mathcal{F}$ on the compact set of points at distance $\le 2r$ from K. So $|f(z) - f(w)| < C|z - w|$ for all $f \in \mathcal{F}$: equicontinuity.

(ii) Let $\{f_n\} \subset \mathcal{F}$ and $K \subset \Omega$ compact. Choose a sequence $\{w_j\}$ dense in Ω. Since $\{f_n\}$ is uniformly bounded, there is a subsequence $\{f_{n,1}\}$ with $f_{n,1}(w_1)$ convergent; from it a subsequence $\{f_{n,2}\}$ with

$f_{n,2}(w_2)$ convergent; and so on. The diagonal sequence $g_n = f_{n,n}$ converges at each w_j. It converges uniformly on K: given $\epsilon > 0$, choose δ as in equicontinuity; K is covered by finitely many discs $D_\delta(w_1), \ldots, D_\delta(w_J)$. Choose N with $|g_m(w_j) - g_n(w_j)| < \epsilon$ for $n, m > N$, $j \le J$. For $z \in K$, $z \in D_\delta(w_j)$ for some $j \le J$, and

$$|g_n(z) - g_m(z)| \le |g_n(z) - g_n(w_j)| + |g_n(w_j) - g_m(w_j)| + |g_m(w_j) - g_m(z)| < 3\epsilon.$$

Finally, with an exhaustion $K_1 \subset K_2 \subset \cdots$, extract $\{g_{n,1}\}$ converging uniformly on K_1, then $\{g_{n,2}\} \subset \{g_{n,1}\}$ on K_2, and so on; $\{g_{n,n}\}$ converges uniformly on every K_ℓ, hence on every compact subset of Ω. ■

Proposition 3.5

If Ω is a connected open subset of $\mathbb{C}$ and $\{f_n\}$ a sequence of injective holomorphic functions on Ω that converges uniformly on every compact subset of Ω to a holomorphic function f, then f is either injective or constant.

Proof. Suppose $f(z_1) = f(z_2)$ with $z_1 \neq z_2$. Let $g_n(z) = f_n(z) - f_n(z_1)$, which has no zero besides z_1, and $g_n \to g = f - f(z_1)$ uniformly on compact sets. If $g \not\equiv 0$, z_2 is an isolated zero of g (Ω is connected), so

$$1 \le \frac{1}{2\pi i} \int_\gamma \frac{g'(\zeta)}{g(\zeta)}\, d\zeta,$$

γ a small circle about z_2 on and inside which g vanishes only at z_2. Then $1/g_n \to 1/g$ and $g_n' \to g'$ uniformly on γ (Theorem 5.3 of Chapter 2), so $\frac{1}{2\pi i}\int_\gamma \frac{g_n'}{g_n} \to \frac{1}{2\pi i}\int_\gamma \frac{g'}{g}$. But g_n has no zeros inside γ, so $\frac{1}{2\pi i}\int_\gamma \frac{g_n'}{g_n} = 0$ for all n, a contradiction. Hence $g \equiv 0$ and f is constant. ■

3.3 Proof of the Riemann mapping theorem

Step 1. Ω is conformally equivalent to an open subset of $\mathbb{D}$ containing 0. Choose $\alpha \notin \Omega$. $z - \alpha$ never vanishes on the simply connected Ω, so there is a holomorphic $f(z) = \log(z - \alpha)$ with $e^{f(z)} = z - \alpha$; in particular f is injective. Fix $w \in \Omega$; then $f(z) \neq f(w) + 2\pi i$ for $z \in \Omega$ (exponentiating would give $z = w$, hence $f(z) = f(w)$). In fact $f(\Omega)$ stays away from $f(w) + 2\pi i$: if $f(z_n) \to f(w) + 2\pi i$, exponentiating gives $z_n \to w$, hence $f(z_n) \to f(w)$, a contradiction. Thus

$$F(z) = \frac{1}{f(z) - (f(w) + 2\pi i)}$$

is injective and bounded on Ω, so $F : \Omega \to F(\Omega)$ is conformal with $F(\Omega)$ bounded; translating and rescaling gives a conformal map of Ω onto an open subset of $\mathbb{D}$ containing 0.

Step 2. So assume $\Omega \subset \mathbb{D}$ is open with $0 \in \Omega$, and let

$$\mathcal{F} = \{f : \Omega \to \mathbb{D} \text{ holomorphic, injective, and } f(0) = 0\}.$$

$\mathcal{F} \neq \emptyset$ (it contains the identity) and is uniformly bounded. The numbers $|f'(0)|$, $f \in \mathcal{F}$, are uniformly bounded by the Cauchy inequalities on a small disc about 0. Let $s = \sup_{f \in \mathcal{F}} |f'(0)|$ and $f_n \in \mathcal{F}$ with $|f_n'(0)| \to s$. By Montel's theorem a subsequence converges uniformly on compact sets to a holomorphic f on Ω. Since $s \ge 1$ (the identity is in $\mathcal{F}$), f is non-constant, hence injective by Proposition 3.5. By continuity $|f(z)| \le 1$, and by the maximum modulus principle $|f(z)| < 1$; also $f(0) = 0$. So $f \in \mathcal{F}$ with $|f'(0)| = s$.

Step 3. f is surjective onto $\mathbb{D}$. Otherwise we construct a function in $\mathcal{F}$ with derivative at 0 greater than s. Suppose $\alpha \in \mathbb{D}$ is not in $f(\Omega)$, and let $\psi_\alpha(z) = \frac{\alpha - z}{1 - \overline{\alpha} z}$. $U = (\psi_\alpha \circ f)(\Omega)$ is simply connected and does not contain 0, so a square root $g(w) = e^{\frac{1}{2}\log w}$ is defined on U. Let

$$F = \psi_{g(\alpha)} \circ g \circ \psi_\alpha \circ f.$$

F is holomorphic, maps 0 to 0 ($g(\psi_\alpha(0)) = g(\alpha)$), maps Ω into $\mathbb{D}$, and is injective (each factor is). So $F \in \mathcal{F}$. With $h(w) = w^2$,

$$f = \psi_\alpha^{-1} \circ h \circ g^{-1} \circ \psi_{g(\alpha)}^{-1} \circ F = \Phi \circ F.$$

$\Phi = \psi_\alpha \circ h \circ \psi_{g(\alpha)}$ maps $\mathbb{D}$ into $\mathbb{D}$ with $\Phi(0) = 0$, and is not injective (since h is not). By the last part of the Schwarz lemma, $|\Phi'(0)| < 1$. Since $f'(0) = \Phi'(0)F'(0)$, $|f'(0)| < |F'(0)|$, contradicting the maximality of $|f'(0)|$.

Finally multiply f by a constant of absolute value 1 so that $f'(0) > 0$. ■

Remark

Simple connectivity was used only for the logarithm and the square root. It would suffice to assume Ω proper and *holomorphically simply connected*: $\int_\gamma f(z)\,dz = 0$ for every holomorphic f in Ω and every closed curve $\gamma \subset \Omega$. See Appendix B.

4 Conformal mappings onto polygons

The Riemann mapping theorem gives little insight into the exact form of the map; but for polygons there are nice formulas: the *Schwarz–Christoffel formula* for conformal maps from the disc (or $\mathbb{H}$) to polygons.

4.1 Some examples

Example 1

For $0 < \alpha < 2$, the map $f(z) = z^\alpha$ of $\mathbb{H}$ onto the sector $\{0 < \arg z < \alpha\pi\}$ (Section 1) can be written

$$z^\alpha = f(z) = \int_0^z f'(\zeta)\,d\zeta = \alpha \int_0^z \zeta^{-\beta}\,d\zeta, \qquad \alpha + \beta = 1,$$

integrated along any path in the (closed) upper half-plane. $\zeta^{-\beta}$ is integrable near 0 since $\beta < 1$, so $f(0) = 0$. For $z = x > 0$, $f'(x) = \alpha x^{\alpha-1} > 0$ and is not integrable at ∞, so f maps $[0, \infty)$ onto $[0, \infty)$. For $x < 0$, $f'(x) = \alpha|x|^{\alpha-1}e^{i\pi(\alpha-1)} = -\alpha|x|^{\alpha-1}e^{i\pi\alpha}$, so f maps $(-\infty, 0]$ onto $(\infty e^{i\pi\alpha}, 0]$.

Example 2

For $z \in \mathbb{H}$ let $f(z) = \int_0^z \frac{d\zeta}{(1-\zeta^2)^{1/2}}$, with the branch of $(1-\zeta^2)^{1/2}$ holomorphic in $\mathbb{H}$ and positive for $-1 < \zeta < 1$; then $(1-\zeta^2)^{-1/2} = -i(\zeta^2-1)^{-1/2}$ for $\zeta > 1$ (and the analogous formula for $\zeta < -1$). f maps $\mathbb{R}$ onto the boundary of the half-strip $\{-\pi/2 < \mathrm{Re} < \pi/2,\ \mathrm{Im} > 0\}$: $f(\pm 1) = \pm\pi/2$ and $f'(x) > 0$ on $(-1, 1)$, so $[-1, 1] \mapsto [-\pi/2, \pi/2]$; for $x > 1$, $f(x) = \frac{\pi}{2} + i\int_1^x \frac{dx}{(x^2-1)^{1/2}}$ and $\int_1^\infty \frac{dx}{(x^2-1)^{1/2}} = \infty$, so $[1, \infty)$ goes onto the vertical half-line above $\pi/2$; similarly for $(-\infty, -1]$. In fact f is the inverse of $\sin z$ (Example 8), so f maps $\mathbb{H}$ conformally onto the half-strip.

Example 3

For fixed $0 < k < 1$, let

$$f(z) = \int_0^z \frac{d\zeta}{[(1-\zeta^2)(1-k^2\zeta^2)]^{1/2}}, \qquad z \in \mathbb{H},$$

the branch positive for $-1 < \zeta < 1$. Such integrals are **elliptic integrals** (variants arise in the arc length of an ellipse). With

$$K = \int_0^1 \frac{dx}{[(1-x^2)(1-k^2x^2)]^{1/2}}, \qquad K' = \int_1^{1/k} \frac{dx}{[(x^2-1)(1-k^2x^2)]^{1/2}},$$

f maps $\mathbb{R}$ onto the rectangle with vertices $-K$, K, $K+iK'$, $-K+iK'$. Divide $\mathbb{R}$ by $-1/k, -1, 1, 1/k$. $f(\pm 1) = \pm K$ and $f' > 0$ on $(-1,1)$, so $[-1,1] \mapsto [-K,K]$. For $1 < x < 1/k$, $f(x) = K + i\int_1^x \frac{d\xi}{[(\xi^2-1)(1-k^2\xi^2)]^{1/2}}$, so $[1,1/k] \mapsto [K, K+iK']$; similarly $[-1/k,-1] \mapsto [-K+iK', -K]$. For $x > 1/k$, $f'(x) = -[(x^2-1)(k^2x^2-1)]^{-1/2}$, so $f(x) = K + iK' - \int_{1/k}^x \frac{dx}{[(x^2-1)(k^2x^2-1)]^{1/2}}$, and since $\int_{1/k}^\infty \frac{dx}{[(x^2-1)(k^2x^2-1)]^{1/2}} = \int_0^1 \frac{dx}{[(1-x^2)(1-k^2x^2)]^{1/2}} = K$ (substitute $x = 1/ku$), $[1/k, \infty)$ goes onto $[K+iK', iK')$; similarly $(-\infty, -1/k]$ onto $[-K+iK', iK')$. The point at infinity goes to the midpoint iK' of the upper side.

Two problems arise: generalize these examples (the Schwarz–Christoffel integral maps $\mathbb{R}$ to a polygonal line), and show that the map is conformal from $\mathbb{H}$ onto the interior of the polygon.

4.2 The Schwarz–Christoffel integral

Definition (Schwarz–Christoffel integral)

For real $A_1 < A_2 < \cdots < A_n$ and exponents $\beta_k < 1$ with $1 < \sum_{k=1}^n \beta_k$, let

$$S(z) = \int_0^z \frac{d\zeta}{(\zeta - A_1)^{\beta_1} \cdots (\zeta - A_n)^{\beta_n}}, \tag{5}$$

where $(z-A_k)^{\beta_k}$ is the branch in the plane slit along $\{A_k + iy : y \le 0\}$ that is positive for real $x > A_k$; thus $(x-A_k)^{\beta_k} = |x-A_k|^{\beta_k} e^{i\pi\beta_k}$ for real $x < A_k$.

The plane slit along the rays $\bigcup_k \{A_k + iy : y \le 0\}$ is simply connected, so S is holomorphic there. Since $\beta_k < 1$, the singularities are integrable, and S is continuous up to $\mathbb{R}$, including the A_k; the integral may be taken along any path avoiding the open slits. Since $\prod_k |\zeta - A_k|^{-\beta_k} \le c|\zeta|^{-\sum \beta_k}$ for large $|\zeta|$ and $\sum \beta_k > 1$, the integral converges at infinity, and by Cauchy's theorem $\lim_{r\to\infty} S(re^{i\theta})$ exists and is independent of $0 \le \theta \le \pi$; call it a_∞, and let $a_k = S(A_k)$. (The cases $\sum \beta_k \le 1$ of Examples 1–2 are excluded; a modification handles them, but then S is unbounded in $\mathbb{H}$.)

Proposition 4.1

Suppose $S(z)$ is given by (5).
(i) If $\sum_{k=1}^n \beta_k = 2$, and p denotes the polygon whose vertices are given (in order) by $a_1, \ldots, a_n$, then S maps the real axis onto $p - \{a_\infty\}$. The point a_∞ lies on the segment $[a_n, a_1]$ and is the image of the point at infinity. Moreover, the (interior) angle at the vertex a_k is $\alpha_k \pi$ where $\alpha_k = 1 - \beta_k$.
(ii) There is a similar conclusion when $1 < \sum_{k=1}^n \beta_k < 2$, except now the image of the extended line is the polygon of $n+1$ sides with vertices $a_1, a_2, \ldots, a_n, a_\infty$. The angle at the vertex a_∞ is $\alpha_\infty \pi$ with $\alpha_\infty = 1 - \beta_\infty$, where $\beta_\infty = 2 - \sum_{k=1}^n \beta_k$.

Proof. Let $\sum \beta_k = 2$. For $A_k < x < A_{k+1}$,

$$S'(x) = \prod_{j \le k} (x - A_j)^{-\beta_j} \prod_{j > k} (x - A_j)^{-\beta_j}, \qquad \arg S'(x) = \arg \prod_{j>k} e^{-i\pi\beta_j} = -\pi \sum_{j>k} \beta_j,$$

constant on (A_k, A_{k+1}). Since $S(x) = S(A_k) + \int_{A_k}^x S'(y)\,dy$, as x goes from A_k to A_{k+1}, $S(x)$ moves along the straight segment $[a_k, a_{k+1}]$ (where $[z,w] = \{(1-t)z + tw : t \in [0,1]\}$), which makes angle

$-\pi\sum_{j>k}\beta_j$ with the real axis. For $x > A_n$, $S'(x) > 0$; for $x < A_1$, $\arg S'(x) = -\pi\sum_k \beta_k = -2\pi$, so $S'(x) > 0$ again. Thus on $[A_n, \infty)$, S moves parallel to the real axis from a_n to a_∞, and on $(-\infty, A_1]$ from a_∞ to a_1; $[a_n, a_\infty) \cup (a_\infty, a_1] = [a_n, a_1] - \{a_\infty\}$. The angle of $[a_k, a_{k+1}]$ exceeds that of $[a_{k-1}, a_k]$ by $\pi\beta_k$, so the interior angle at a_k is $\pi - \pi\beta_k = \alpha_k\pi$. The case $1 < \sum \beta_k < 2$ is similar. ■

This does not settle the problem:

(1) for general n and A_k the image polygon need not be simple, nor S conformal on $\mathbb{H}$;

(2) it does not show that for a given simply connected region P bounded by a polygon, S (with suitable A_k and simple modifications) is a conformal map of $\mathbb{H}$ onto P. That is the case, as we now prove.

4.3 Boundary behavior

A **polygonal region** P is a bounded simply connected open set whose boundary is a closed polygonal line p (a *polygon*). For a conformal $f : U \to f(U)$,

$$\text{Area}(f(U)) = \iint_U |f'(z)|^2\, dx\, dy,$$

since the Jacobian of $w = f(z)$ is $|f'(z)|^2$ (Proposition 2.3 of Chapter 1).

Theorem 4.2

If $F : \mathbb{D} \to P$ is a conformal map, then F extends to a continuous bijection from the closure $\overline{\mathbb{D}}$ of the disc to the closure $\overline{P}$ of the polygonal region. In particular, F gives rise to a bijection from the boundary of the disc to the boundary polygon p.

Lemma 4.3

For each $0 < r < 1/2$, let C_r denote the circle centered at z_0 (a point of the unit circle) of radius r. Suppose that for all sufficiently small r we are given two points z_r and z_r' in the unit disc that also lie on C_r. If we let $\rho(r) = |F(z_r) - F(z_r')|$, then there exists a sequence $\{r_n\}$ of radii that tends to zero, and such that $\lim_{n\to\infty} \rho(r_n) = 0$.

Proof. If not, there are $c > 0$, $0 < R < 1/2$ with $c \le \rho(r)$ for $0 < r \le R$. Now $F(z_r) - F(z_r') = \int_\gamma F'(\zeta)\, d\zeta$, γ the arc of C_r in $\mathbb{D}$ joining z_r, z_r'; parametrizing it by $z_0 + re^{i\theta}$, $\theta_1(r) \le \theta \le \theta_2(r)$,

$$\rho(r) \le \int_{\theta_1(r)}^{\theta_2(r)} |F'(z)| r\, d\theta \le \Big(\int_{\theta_1(r)}^{\theta_2(r)} |F'(z)|^2 r\, d\theta\Big)^{1/2} \Big(\int_{\theta_1(r)}^{\theta_2(r)} r\, d\theta\Big)^{1/2}$$

by Cauchy–Schwarz. Squaring and dividing by r, $\frac{\rho(r)^2}{r} \le 2\pi \int_{\theta_1(r)}^{\theta_2(r)} |F'(z)|^2 r\, d\theta$. Integrating from 0 to R,

$$c^2 \int_0^R \frac{dr}{r} \le 2\pi \int_0^R \int_{\theta_1(r)}^{\theta_2(r)} |F'(z)|^2 r\, d\theta\, dr \le 2\pi \iint_{\mathbb{D}} |F'(z)|^2\, dx\, dy = 2\pi\, \text{Area}(P).$$

The left side is infinite and the right side finite—a contradiction. ■

Lemma 4.4

Let z_0 be a point on the unit circle. Then $F(z)$ tends to a limit as z approaches z_0 within the unit disc.

Proof. If not, there are sequences $z_n, z_n' \to z_0$ in $\mathbb{D}$ with $F(z_n) \to \zeta$, $F(z_n') \to \zeta'$, distinct points of $\overline{P}$; since F is conformal, they lie on p. Choose disjoint discs D, D' about ζ, ζ' at distance $d > 0$.

For large n, $F(z_n) \in D$, $F(z_n') \in D'$, so there are continuous curves $\Lambda \subset D \cap P$, $\Lambda' \subset D' \cap P$ (images of continuous maps of a closed interval) containing $F(z_n)$, $F(z_n')$ for all large n, with end-points ζ, ζ'. Then $\lambda = F^{-1}(\Lambda)$ and $\lambda' = F^{-1}(\Lambda')$ are continuous curves in $\mathbb{D}$, containing infinitely many z_n, resp. z_n', which tend to z_0. By continuity, for all small r the circle C_r meets λ and λ' at points z_r, z_r', and $|F(z_r) - F(z_r')| > d$, contradicting Lemma 4.3. ■

Lemma 4.5

The conformal map F extends to a continuous function from the closure of the disc to the closure of the polygon.

Proof. Define $F(z_0) = \lim_{z \to z_0} F(z)$ for $|z_0| = 1$ (Lemma 4.4). Given ϵ, there is δ with $|F(z) - F(z_0)| < \epsilon$ for $z \in \mathbb{D}$, $|z - z_0| < \delta$. If $|z| = 1$ and $|z - z_0| < \delta$, choose $w \in \mathbb{D}$ with $|F(z) - F(w)| < \epsilon$ and $|w - z_0| < \delta$; then $|F(z) - F(z_0)| \le |F(z) - F(w)| + |F(w) - F(z_0)| < 2\epsilon$. ■

Proof of Theorem 4.2. F extends continuously to $\overline{\mathbb{D}} \to \overline{P}$. The argument applies to the inverse G of F: the key geometric property used was that for $z_0 \in \partial\mathbb{D}$ and a small circle C about z_0, $C \cap \mathbb{D}$ is an arc; this holds at every boundary point of P. So G extends continuously to $\overline{P} \to \overline{\mathbb{D}}$. If $z \in \overline{\mathbb{D}}$ and $z_k \to z$ in $\mathbb{D}$, then $G(F(z_k)) = z_k$, and by continuity $G(F(z)) = z$; similarly $F(G(w)) = w$ for $w \in \overline{P}$. ■

4.4 The mapping formula

Let P be a polygonal region bounded by p with consecutive vertices $a_1, \dots, a_n$, $n \ge 3$, interior angle $\alpha_k \pi$ at a_k, and exterior angle $\beta_k \pi$ with $\alpha_k + \beta_k = 1$; a simple geometric argument gives $\sum_{k=1}^{n} \beta_k = 2$. The correspondence $w = (i - z)/(i + z)$ between $\mathbb{H}$ and $\mathbb{D}$ sends the point at infinity to $w = -1$, so a conformal map of $\mathbb{H}$ onto P (which exists by the Riemann mapping theorem) extends to a continuous bijection of $\mathbb{R} \cup \{\infty\}$ onto p. Let F be such a map, and first assume no vertex corresponds to ∞: $F(A_k) = a_k$ for real A_k. Since F is continuous and injective and the vertices are consecutive, after relabeling $A_1 < A_2 < \cdots < A_n$, and F maps $[A_k, A_{k+1}]$ onto $[a_k, a_{k+1}]$ and $(-\infty, A_1] \cup [A_n, \infty)$ onto $[a_n, a_1]$.

Theorem 4.6 (Schwarz–Christoffel)

There exist complex numbers c_1 and c_2 so that the conformal map F of $\mathbb{H}$ to P is given by $F(z) = c_1 S(z) + c_2$, where S is the Schwarz–Christoffel integral introduced in Section 4.2.

Proof. Let $1 < k < n$. F maps $[A_{k-1}, A_k]$ and $[A_k, A_{k+1}]$ onto two segments meeting at $a_k = F(A_k)$ at angle $\alpha_k \pi$. Choosing a branch of the logarithm, define

$$h_k(z) = (F(z) - a_k)^{1/\alpha_k}$$

on the half-strip in $\mathbb{H}$ between $\mathrm{Re}(z) = A_{k-1}$ and $\mathrm{Re}(z) = A_{k+1}$. h_k is continuous up to (A_{k-1}, A_{k+1}) and maps $[A_{k-1}, A_{k+1}]$ onto a straight segment L_k, with $A_k \mapsto 0$. By the Schwarz reflection principle, h_k continues analytically to the whole strip $A_{k-1} < \mathrm{Re}(z) < A_{k+1}$. h_k' never vanishes there: in the upper half-strip,

$$\frac{F'(z)}{F(z) - F(A_k)} = \frac{1}{\alpha_k} \frac{h_k'(z)}{h_k(z)},$$

and $F'(z) \neq 0$ (Proposition 1.1), so $h_k'(z) \neq 0$; by reflection the same holds in the lower half-strip. For $A_{k-1} < x < A_{k+1}$, the image under h_k of a small half-disc about x in $\mathbb{H}$ lies on one side of L_k; h_k is injective up to L_k (since F is), and by the symmetry of reflection h_k is injective on the whole disc about x, so $h_k'(x) \neq 0$.

Since $F = a_k + h_k^{\alpha_k}$, $F' = \alpha_k h_k^{\alpha_k - 1} h_k'$ and $F'' = \alpha_k(\alpha_k - 1) h_k^{\alpha_k - 2} (h_k')^2 + \alpha_k h_k^{\alpha_k - 1} h_k''$; as $h_k' \neq 0$ and

h_k has a simple zero at A_k,

$$\frac{F''(z)}{F'(z)} = \frac{-\beta_k}{z - A_k} + E_k(z),$$

E_k holomorphic in the strip $A_{k-1} < \mathrm{Re}(z) < A_{k+1}$. Similarly $\frac{F''}{F'} = \frac{-\beta_1}{z-A_1} + E_1$ in $-\infty < \mathrm{Re}(z) < A_2$ and $\frac{F''}{F'} = \frac{-\beta_n}{z-A_n} + E_n$ in $A_{n-1} < \mathrm{Re}(z) < \infty$. Moreover F continues across $(-\infty, A_1) \cup (A_n, \infty)$ by reflection (the image is a straight segment), so F continues to $|z| > R$ for large R; since F maps $\mathbb{H}$ into a bounded region, the continuation is bounded, hence holomorphic at infinity: $F(z) = c_0 + \frac{c_1}{z} + \frac{c_2}{z^2} + \cdots$. Differentiating, F''/F' decays like $1/z$ at infinity. Since the strips overlap and cover $\mathbb{C}$,

$$\frac{F''(z)}{F'(z)} + \sum_{k=1}^{n} \frac{\beta_k}{z - A_k}$$

is entire and vanishes at infinity; by Liouville it is zero. With $Q(z) = (z - A_1)^{-\beta_1} \cdots (z - A_n)^{-\beta_n}$, $\frac{Q'}{Q} = -\sum_k \frac{\beta_k}{z-A_k} = \frac{F''}{F'}$, so $\frac{d}{dz}\left(\frac{F'(z)}{Q(z)}\right) = 0$ and $F' = cQ$. A final integration yields the theorem. ■

Theorem 4.7

If F is a conformal map from the upper half-plane to the polygonal region P and maps the points $A_1, \ldots, A_{n-1}, \infty$ to the vertices of p, then there exist constants C_1 and C_2 such that

$$F(z) = C_1 \int_0^z \frac{d\zeta}{(\zeta - A_1)^{\beta_1} \cdots (\zeta - A_{n-1})^{\beta_{n-1}}} + C_2.$$

So the formula is obtained by deleting the last factor of the Schwarz–Christoffel integral.

Proof. After a translation assume $A_j \neq 0$ for $j = 1, \ldots, n-1$. Choose a real $A_n^* > 0$ (supplied: different from the others) and let $\Phi(z) = A_n^* - \frac{1}{z}$, an automorphism of $\mathbb{H}$. Let $A_k^* = \Phi(A_k)$, $k \leq n-1$, and note $A_n^* = \Phi(\infty)$. Then $(F \circ \Phi^{-1})(A_k^*) = a_k$ for $k = 1, \ldots, n$, and by Theorem 4.6

$$(F \circ \Phi^{-1})(z') = C_1' \int_0^{z'} \frac{d\zeta}{(\zeta - A_1^*)^{\beta_1} \cdots (\zeta - A_n^*)^{\beta_n}} + C_2'.$$

Change variables $\zeta = \Phi(w)$, $d\zeta = dw/w^2$; since $\zeta - A_n^* = -1/w$ and $\zeta - A_k^* = (w(A_n^* - A_k^*) - 1)/w$, and $2 = \beta_1 + \cdots + \beta_n$, the integrand becomes $C \prod_{k<n}(w - 1/(A_n^* - A_k^*))^{-\beta_k}$ up to a constant factor. Since $1/(A_n^* - A_k^*) = A_k$, setting $\Phi^{-1}(z') = z$ gives the formula. ■

4.5 Return to elliptic integrals

Consider again $I(z) = \int_0^z \frac{d\zeta}{[(1-\zeta^2)(1-k^2\zeta^2)]^{1/2}}$, $0 < k < 1$, which maps $\mathbb{R}$ onto the rectangle R with vertices $-K, K, K+iK', -K+iK'$ (Example 3). We show I is a conformal map of $\mathbb{H}$ onto the interior of R. By Theorem 4.6 there is a conformal map F of $\mathbb{H}$ onto R mapping four real points to the vertices; composing with an automorphism of $\mathbb{H}$ we may assume F maps $-1, 0, 1$ to $-K, 0, K$. Let $1/\ell$ $(0 < \ell < 1)$ be the real point mapped to $K + iK'$. Then $F(-1/\ell) = -K + iK'$: $\tilde{F}(z) = -\overline{F(-\bar{z})}$ is, by the symmetry of R, also a conformal map of $\mathbb{H}$ onto R with $\tilde{F}(0) = 0$, $\tilde{F}(\pm 1) = \pm K$; so $F^{-1} \circ \tilde{F}$ is an automorphism of $\mathbb{H}$ fixing $-1, 0, 1$, hence the identity, and $F(-1/\ell) = -\overline{F(1/\ell)} = -K + iK'$. By Theorem 4.6,

$$F(z) = c_1 \int_0^z \frac{d\zeta}{[(1-\zeta^2)(1-\ell^2\zeta^2)]^{1/2}} + c_2.$$

$z = 0$ gives $c_2 = 0$; $z = 1$ and $z = 1/\ell$ give $K(k) = c_1 K(\ell)$ and $K'(k) = c_1 K'(\ell)$, where $K(k) = \int_0^1 \frac{dx}{[(1-x^2)(1-k^2x^2)]^{1/2}}$, $K'(k) = \int_1^{1/k} \frac{dx}{[(x^2-1)(1-k^2x^2)]^{1/2}}$. $K(k)$ is strictly increasing in $k \in (0,1)$, and a change of variables gives $K'(k) = K(\tilde{k})$ with $\tilde{k}^2 = 1 - k^2$, $\tilde{k} > 0$, so K' is strictly decreasing. Hence $K(k)/K'(k)$ is strictly increasing; since $K(k)/K'(k) = K(\ell)/K'(\ell)$, $k = \ell$ and $c_1 = 1$. Thus $I = F$ is conformal.

Passing to the inverse function gives basic insight: let $z \mapsto \mathrm{sn}(z)$ be the inverse of $z \mapsto I(z)$ (Jacobi's notation, by analogy with $\sin z$). It maps the closed rectangle $R_0 = R$ onto the closed upper half-plane. Reflect successively along the lower sides to get rectangles $R_1, R_2, \ldots$: by the reflection principle, $\mathrm{sn}(z) = \overline{\mathrm{sn}(\overline{z})}$ for $z \in R_1$, then $\mathrm{sn}(z) = \mathrm{sn}(-iK' + \overline{z})$ for $z \in R_2$, and so on; this extends sn to the strip $-K < \mathrm{Re}(z) < K$ with $\mathrm{sn}(z) = \mathrm{sn}(z + 2iK')$. Reflecting also in horizontal rows of rectangles, sn continues to all of $\mathbb{C}$ with $\mathrm{sn}(z) = \mathrm{sn}(z+4K)$. Thus sn is *doubly periodic*, with periods $4K$ and $2iK'$ (and meromorphic, with two simple poles in a period parallelogram)—the subject of Chapter 9.